

WEST VANCOUVER, BC, Canada

WMO: 717840

Lat: **49.3470N**Lon: **123.1933W**Elev: **558**StdP: **14.40**Time Zone: **-8.00 (NAP)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	22.7	26.6	10.0	9.4	27.9	16.5	12.9	29.1	13.7	42.0	12.1	42.0	2.1	10	0.453

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	15.7	81.1	64.4	77.8	63.9	74.8	62.7	67.0	77.3	65.3	75.0	63.8	72.9	2.1	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.8	87.4	72.3	61.2	82.5	70.2	59.9	78.5	68.4	31.7	77.5	30.4	75.2	29.3	73.2	73.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.3	8.8	7.4	DB	19.3	88.7	5.0	4.1	15.7	91.6	12.8	94.1	10.0	96.4	6.4	99.4	(4)
(5)			WB	17.9	69.0	4.8	1.8	14.4	70.3	11.6	71.4	8.9	72.4	5.4	73.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.9	39.7	40.8	43.9	48.6	55.1	59.7	64.8	65.2	59.9	50.8	43.6	38.5
	DBStd	10.81	6.00	5.12	4.85	5.01	5.64	5.56	5.08	4.49	5.26	4.69	5.58	5.87
	HDD50	1483	321	259	197	84	15	1	0	0	0	46	202	358
	HDD65	5288	785	678	654	492	310	178	64	52	171	439	643	822
	CDD50	1826	1	1	8	41	173	293	458	472	297	72	9	1
	CDD65	157	0	0	0	0	4	21	57	58	17	0	0	0
	CDH74	828	0	0	0	3	29	139	317	265	75	0	0	0
	CDH80	153	0	0	0	0	3	31	71	39	9	0	0	0
Wind	WSAvg	2.6	2.9	2.8	3.1	2.8	2.4	2.4	2.2	2.2	2.2	2.6	3.0	3.0
Precipitation	PrecAvg	88.3	12.5	8.1	8.5	6.0	4.1	3.4	2.1	2.8	4.2	10.0	13.7	12.8
	PrecMax	111.6	20.8	16.1	15.8	12.4	6.8	6.6	5.0	12.6	9.2	17.1	25.7	18.5
	PrecMin	70.1	7.0	0.4	2.0	2.2	1.0	1.0	0.5	0.4	0.6	1.6	6.2	7.9
	PrecStd	10.0	3.3	3.2	2.8	2.3	1.4	1.6	1.2	2.6	2.3	3.7	4.8	2.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.5	55.6	61.5	70.6	78.0	84.4	87.2	84.2	80.7	67.5	58.8	54.5
		MCWB	49.9	47.7	51.0	55.7	59.1	64.4	68.1	65.4	62.7	57.1	53.1	49.4
	2%	DB	51.7	52.0	57.1	64.7	72.3	78.1	80.9	79.9	75.6	63.0	55.4	50.7
		MCWB	48.5	46.4	48.6	53.5	58.8	62.8	65.4	64.8	61.7	55.1	52.0	47.4
	5%	DB	49.5	49.7	53.7	60.1	68.3	72.9	77.1	76.7	71.6	60.0	53.1	48.1
		MCWB	47.0	45.4	46.9	50.7	57.0	61.3	64.3	64.4	60.9	53.9	50.3	45.8
	10%	DB	47.3	47.7	51.0	56.6	64.7	69.3	74.0	73.9	68.1	57.4	51.0	46.0
		MCWB	45.1	44.1	46.0	48.9	55.7	59.5	63.0	63.1	59.0	52.6	48.5	44.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.2	50.8	53.2	58.4	62.6	66.4	70.2	68.4	65.8	59.9	54.9	51.3
		MCDB	53.0	53.3	57.6	67.0	72.6	78.9	82.0	78.3	76.3	64.0	56.3	52.6
	2%	WB	49.5	48.1	50.5	54.3	60.0	64.1	67.6	66.6	63.6	57.2	52.9	48.2
		MCDB	50.9	50.5	54.7	62.2	69.0	75.3	78.0	76.5	72.4	60.6	54.7	49.9
	5%	WB	47.4	46.2	48.4	52.0	58.1	62.0	65.6	65.2	61.8	55.3	50.9	46.1
		MCDB	48.9	48.8	52.0	58.5	66.1	71.8	74.9	74.6	69.4	58.4	53.0	47.7
	10%	WB	45.5	44.6	46.7	50.0	56.3	60.0	63.8	63.7	60.0	53.3	48.9	44.3
		MCDB	46.9	47.0	50.0	55.1	63.7	68.3	72.5	72.6	66.6	56.7	50.7	45.8
Mean Daily Temperature Range	5% DB	MDBR	8.2	10.6	11.8	14.1	15.2	15.0	16.0	15.7	14.3	11.0	8.9	7.7
		MCDBR	10.2	13.6	15.4	19.2	20.3	20.7	20.5	19.6	18.8	14.3	10.7	9.3
		MCWBR	7.0	8.9	9.2	11.0	11.2	11.0	10.5	10.2	10.0	8.7	7.4	6.8
	5% WB	MCDBR	9.1	11.1	13.6	17.3	18.1	19.5	19.3	18.5	17.0	12.8	9.0	8.6
		MCWBR	7.0	7.8	8.9	10.5	10.6	10.9	10.8	10.3	9.6	8.5	6.9	7.0
Clear-Sky Solar Irradiance	taub		0.311	0.304	0.323	0.355	0.376	0.357	0.354	0.357	0.340	0.330	0.314	0.299
	taud		2.417	2.481	2.451	2.389	2.375	2.454	2.486	2.493	2.543	2.552	2.505	2.443
	Ebn at Noon		235	271	283	283	280	286	285	280	274	258	234	224
	Edh at Noon		20	24	29	35	36	34	33	31	27	23	19	17
All-Sky Solar Radiation	RadAvg		267	538	799	1252	1647	1760	1978	1680	1157	635	313	213
	RadStd		46	84	114	140	146	220	182	137	106	92	61	34